MACHINE LEARNING

UNIT 1 – Machine Learning Foundations and Statistical Learning

Exam-Oriented Theory Notes + Important Questions + Revision Material

Syllabus Area	Coverage in this PDF
Foundations	Introduction to ML, AI vs ML vs DL, types of learning
ML Workflow	ML pipeline, preprocessing, feature engineering, EDA
Statistical Learning	Overfitting, bias–variance, train/validation/test
Evaluation	Evaluation strategies and common metrics
Case Study	Customer retention prediction using behavioral analytics
Practical Focus	Preprocessing, features, EDA, pipeline, training-validation

★ Questions marked with a red/yellow box are high-priority exam questions. Learn their definitions, differences, steps, examples, and diagrams.

How to use: First read the theory once, then revise the comparison tables, then practice the Important Questions and finally use the one-page revision checklist at the end.

1. Unit 1 Syllabus Map

The given syllabus contains the complete foundation of a typical machine-learning workflow. The most exam-relevant areas are conceptual definitions, comparisons, process diagrams, bias–variance, data splitting, preprocessing, feature engineering, and evaluation.

Topic	What to prepare for exam	Priority
Introduction to ML	Definition, goals, applications, examples	★★★
AI vs ML vs DL	Definitions + hierarchy + differences + examples	★★★★★
Types of Learning	Supervised, unsupervised, semi-supervised, reinforcement	★★★★★
ML Pipeline	All stages and their purpose; diagram	★★★★★
Data Preprocessing	Missing values, encoding, scaling, outliers, duplicates	★★★★★
Feature Engineering	Feature creation, selection, transformation, extraction	★★★★★
EDA	Purpose, statistics, plots, correlation, outliers	★★★★
Overfitting & Bias–Variance	Definitions, causes, symptoms, remedies	★★★★★
Train/Validation/Test	Purpose and common splits; leakage	★★★★★
Evaluation Strategies	Cross-validation + classification/regression metrics	★★★★★
Customer Retention Case	Problem formulation, features, model, metrics, actions	★★★★

Priority guide: ★★★★★ = very likely / high-value; ★★★★ = important; ★★★ = basic but should be prepared.

2. Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence in which computers learn patterns from data and use those learned patterns to make predictions, classifications, recommendations, or decisions without being explicitly programmed with a separate rule for every case.

Why do we need Machine Learning?

• Traditional programming requires explicit rules for many situations. ML is useful when the rules are complex, numerous, or difficult to write manually.

• ML can discover statistical patterns from historical data and apply them to unseen data.

• It is widely used in recommendation systems, fraud detection, spam filtering, medical decision support, forecasting, search, computer vision and natural language processing.

Basic ML terminology

Term	Meaning
Dataset	Collection of observations/examples used for analysis or model development.
Feature	Input variable used by a model, e.g., age, income, number of logins.
Target / Label	Output variable the model tries to predict in supervised learning.
Model	Learned mathematical/computational relationship between inputs and output.
Training	Process of learning model parameters from training data.
Inference / Prediction	Using a trained model on new/unseen input data.
Parameter	Value learned internally by the model during training.

Term	Meaning
Hyperparameter	Configuration chosen before/during training, e.g., tree depth or learning rate.

★ IMPORTANT EXAM QUESTION: Define Machine Learning. Explain its need, characteristics and real-world applications with examples.

★ IMPORTANT EXAM QUESTION: Differentiate traditional programming and machine learning.

Exam Tip: For a 5–10 mark answer, use: definition → working idea → components → applications → short example.

3. AI vs ML vs Deep Learning

Artificial Intelligence (AI)

AI is the broad field concerned with building systems that perform tasks associated with human intelligence, such as reasoning, planning, perception, language understanding and decision-making.

Machine Learning (ML)

ML is a subset of AI that learns patterns from data. Instead of specifying every rule, we provide examples and an algorithm learns a model.

Deep Learning (DL)

Deep Learning is a subset of ML based mainly on multi-layer neural networks. It is particularly effective for high-dimensional data such as images, audio, video and natural language.

Aspect	AI	ML	DL
Scope	Broadest field	Subset of AI	Subset of ML
Main idea	Intelligent behavior	Learning from data	Learning complex representations using deep neural networks
Data need	Can use rules, knowledge or data	Usually needs data	Usually benefits from large datasets
Feature engineering	May be manual/rule-based	Often important	Often learned automatically
Example	Planning system	Spam classifier	Image recognition with CNN

Hierarchy to remember: **AI** ⊃ **ML** ⊃ **DL**. Deep Learning is not separate from ML; it is a specialized approach within ML.

★ IMPORTANT EXAM QUESTION: Differentiate AI, ML and DL with a suitable diagram and examples.

★ IMPORTANT EXAM QUESTION: Explain the relationship between Artificial Intelligence, Machine Learning and Deep Learning.

4. Types of Machine Learning

4.1 Supervised Learning

In supervised learning, the training dataset contains input features and known target labels. The model learns a mapping from inputs X to an output y.

Examples: house-price prediction, disease classification, spam detection, customer churn prediction.

Main supervised tasks

Task	Output	Example	Common algorithms
Classification	Discrete class/category	Spam / Not Spam	Logistic Regression, Decision Tree, Random Forest, SVM, KNN
Regression	Continuous numerical value	House price	Linear Regression, Decision Tree Regression, Random Forest Regression

4.2 Unsupervised Learning

In unsupervised learning, the data does not have target labels. The algorithm tries to discover hidden structure or patterns in the input data.

• **Clustering:** groups similar observations, e.g., customer segmentation.

• **Dimensionality reduction:** represents high-dimensional data with fewer dimensions while preserving useful information, e.g., PCA.

• **Association analysis:** finds relationships among items, e.g., products frequently purchased together.

4.3 Semi-Supervised Learning

Semi-supervised learning uses a relatively small amount of labeled data together with a larger amount of unlabeled data. It is useful when labeling is expensive but collecting raw data is easy.

4.4 Reinforcement Learning

In reinforcement learning, an agent interacts with an environment, takes actions, receives rewards or penalties, and learns a policy that aims to maximize cumulative reward.

Type	Training signal	Typical example
Supervised	Known labels	Email spam classification
Unsupervised	No labels; discover structure	Customer clustering
Semi-supervised	Few labels + many unlabeled examples	Image classification with limited labeling
Reinforcement	Rewards/penalties from environment	Game-playing agent / robot control

★ IMPORTANT EXAM QUESTION: Explain supervised and unsupervised learning with examples. Differentiate them.

★ IMPORTANT EXAM QUESTION: Explain the different types of machine learning.

★ IMPORTANT EXAM QUESTION: Differentiate classification and regression.

★ IMPORTANT EXAM QUESTION: What is reinforcement learning? Explain agent, environment, action, reward and policy.

5. Machine Learning Pipeline

An ML pipeline is a structured sequence of steps that converts raw data into a trained and evaluated model and finally into predictions. A pipeline improves reproducibility, reduces errors and helps prevent data leakage.

Stage	Purpose
1. Problem definition	Define business/scientific objective, target and success criteria.
2. Data collection	Collect relevant historical or real-time data.
3. Data understanding / EDA	Inspect distributions, missingness, relationships, outliers and class balance.
4. Data preprocessing	Clean, encode, scale and transform data.

Stage	Purpose
5. Feature engineering	Create/select/transform informative input variables.
6. Data splitting	Create training, validation and test sets appropriately.
7. Model selection	Choose algorithms suitable for the problem and data.
8. Training	Fit model parameters using training data.
9. Validation / tuning	Tune hyperparameters and compare candidate models.
10. Final testing	Estimate generalization performance on untouched test data.
11. Deployment	Make the model available for real-world inference.
12. Monitoring	Track drift, errors, latency and data/model changes after deployment.

Exam flow: Problem → Data → EDA → Preprocessing → Features → Split → Train → Validate/Tune → Test → Deploy → Monitor

★ IMPORTANT EXAM QUESTION: Explain the complete Machine Learning pipeline with a neat diagram.

★ IMPORTANT EXAM QUESTION: Why is data preprocessing performed before model training? Explain the major steps.

★ IMPORTANT EXAM QUESTION: What is data leakage? Explain how it can occur in an ML pipeline and how to prevent it.

Exam Tip: Draw the pipeline as boxes with arrows. Under each box write one purpose. This can turn a short answer into a strong long answer.

6. Data Preprocessing

Data preprocessing converts raw, inconsistent or incomplete data into a form suitable for machine-learning algorithms. Poor-quality input data can produce unreliable models even when the algorithm itself is appropriate.

6.1 Handling Missing Values

• Identify missing values using null/NA checks and summary statistics.

• Numerical data: mean or median imputation can be used depending on distribution and outliers.

• Categorical data: mode or a separate 'Unknown' category may be used.

• For substantial missingness, a feature/row may sometimes be removed if justified.

• Imputation values must be learned from training data only when building a predictive pipeline.

6.2 Encoding Categorical Variables

Method	Idea	Suitable use
Label / Ordinal Encoding	Map categories to integers	Ordered categories or tree-based contexts when appropriate
One-Hot Encoding	Create binary column per category	Nominal categories
Target Encoding	Encode using target-related statistics	High-cardinality categories; must prevent leakage

6.3 Feature Scaling

Scaling puts numerical features on comparable ranges. It is especially important for distance-based and gradient-based algorithms.

Method	Formula / idea	Notes
Standardization	$z = (x - \mu) / \sigma$	Mean approximately 0 and standard deviation approximately 1.
Min-Max Scaling	$x' = (x - \min)/(\max - \min)$	Maps values typically to [0,1]. Sensitive to outliers.
Robust Scaling	Uses median and IQR	More resistant to extreme outliers.

6.4 Outliers

An outlier is an observation that is unusually far from the general pattern of the data. Outliers can be genuine observations, measurement errors, or rare events.

• Detection: box plot, IQR rule, z-scores, domain-specific limits.

• Treatment: verify the data, cap/winsorize, transform, use robust methods, or remove only when justified.

• Do not automatically delete every outlier; some outliers contain important information.

6.5 Duplicates and Inconsistent Data

Remove accidental duplicate records and standardize inconsistent formats such as spelling, units, date formats and category names.

★ IMPORTANT EXAM QUESTION: Explain data preprocessing and discuss missing-value handling, encoding, scaling and outlier treatment.

★ IMPORTANT EXAM QUESTION: What is feature scaling? Compare standardization and min-max normalization.

★ IMPORTANT EXAM QUESTION: Explain one-hot encoding with an example.

7. Feature Engineering

Feature engineering is the process of creating, transforming, selecting or extracting input variables so that a machine-learning model can learn useful patterns more effectively.

Types of Feature Engineering

• **Feature creation:** derive new variables from existing variables. Example: total_spend = price × quantity.

• **Feature transformation:** apply log, square-root, binning or other transformations to improve representation.

• **Feature selection:** keep informative features and remove irrelevant/redundant ones.

• **Feature extraction:** transform many original variables into a smaller set of representations, e.g., PCA.

• **Domain-based features:** use subject knowledge, e.g., recency, frequency and monetary value for customer behavior.

Good feature engineering should

• Improve signal-to-noise ratio.

• Represent information in a form the chosen algorithm can use effectively.

• Avoid leakage from future or target information.

• Be reproducible: the same transformations must be applied at training and inference time.

Example: Customer Retention

Suppose a company wants to predict whether a customer will leave. Raw data may contain transaction records and login logs. Useful engineered features can include **days_since_last_login**, **number_of_logins_last_30_days**,

average_order_value, **support_tickets_last_90_days** and **percentage_change_in_usage**.

★ IMPORTANT EXAM QUESTION: What is feature engineering? Explain feature creation, transformation, selection and extraction with examples.

★ IMPORTANT EXAM QUESTION: Why is feature engineering important in machine learning?

★ IMPORTANT EXAM QUESTION: Design suitable features for a customer retention/churn prediction problem.

Exam Tip: Remember: **Feature engineering = make raw data more meaningful for the model.**

8. Exploratory Data Analysis (EDA)

EDA is the systematic examination of a dataset before or during modeling to understand its structure, quality, distributions, relationships, anomalies and potential patterns.

EDA Checklist

• Inspect number of rows and columns and identify data types.

• Check missing values and duplicates.

• Calculate descriptive statistics: mean, median, mode, variance, standard deviation, quartiles.

• Inspect distributions using histograms or density plots.

• Use box plots to identify possible outliers.

• Study relationships using scatter plots and correlation matrices.

• For categorical variables, inspect frequencies and class imbalance.

• Compare important features with the target variable.

Descriptive Statistics

Measure	Meaning
Mean	Arithmetic average; sensitive to extreme values.
Median	Middle value after sorting; more robust to outliers.
Mode	Most frequent value.
Variance	Average squared deviation from the mean.
Standard deviation	Square root of variance; measures dispersion in original units.
Quartiles	Values dividing ordered data into four parts.
IQR	Q3 – Q1; commonly used in outlier detection.

★ IMPORTANT EXAM QUESTION: What is EDA? Explain its objectives, steps and commonly used plots.

★ IMPORTANT EXAM QUESTION: Differentiate mean and median. Which is more robust to outliers and why?

★ IMPORTANT EXAM QUESTION: Explain how correlation analysis can be used during EDA.

9. Overfitting, Underfitting and Bias–Variance

Overfitting

Overfitting occurs when a model learns the training data too closely, including noise or accidental patterns, and therefore performs well on training data but poorly on unseen data.

Underfitting

Underfitting occurs when a model is too simple to capture important patterns. It tends to perform poorly on both training and validation/test data.

Situation	Training error	Validation/Test error	Typical cause
Underfitting	High	High	Model too simple / insufficient learning
Good fit	Low	Low and similar	Appropriate complexity and generalization
Overfitting	Very low	High	Model too complex, noisy data, insufficient regularization

Bias–Variance Trade-off

Bias is error caused by overly restrictive assumptions in the model. High bias usually means the model is too simple and misses important structure.

Variance is error caused by a model being overly sensitive to the particular training sample. High variance is associated with overfitting.

The bias–variance trade-off describes the need to balance under-complex and over-complex models. Increasing model complexity can reduce bias but may increase variance.

Ways to reduce overfitting

• Use more representative training data.

• Reduce model complexity.

• Apply regularization such as L1 or L2 where appropriate.

• Use cross-validation for reliable model selection.

• Perform feature selection or dimensionality reduction when justified.

• Use early stopping for suitable iterative models.

• Use ensemble methods such as bagging when appropriate.

★ IMPORTANT EXAM QUESTION: What is overfitting? Explain its causes, symptoms and methods to prevent it.

★ IMPORTANT EXAM QUESTION: Explain underfitting and overfitting with training and validation error.

★ IMPORTANT EXAM QUESTION: Explain the bias–variance trade-off with a suitable diagram.

Exam Tip: A common diagram is: model complexity on x-axis; error on y-axis. Training error generally decreases, while validation error often decreases initially and then rises after overfitting begins.

10. Train, Validation and Test Split

Dataset splitting is used to estimate how well a model generalizes to unseen data and to prevent model selection from being based on the final evaluation set.

Subset	Purpose	Used when
Training set	Learn model parameters	Model fitting
Validation set	Tune hyperparameters, choose models, compare configurations	During development
Test set	Final unbiased-ish estimate of generalization after decisions are complete	At the end

A common split is **70:15:15** or **80:10:10**, but the appropriate split depends on dataset size, task and validation strategy. In cross-validation, a separate fixed validation set may not be necessary for every experiment.

Data Leakage

Data leakage occurs when information that would not be available at prediction time, or information derived from the validation/test set, influences training or model selection. Leakage can produce unrealistically strong evaluation results.

• Fit preprocessing transformations using training data and then apply them to validation/test data.

• Do not use future information to construct past prediction features.

• Keep the final test set untouched until final evaluation.

★ IMPORTANT EXAM QUESTION: Differentiate training, validation and test datasets.

★ IMPORTANT EXAM QUESTION: Why should the test set not be used repeatedly for model tuning?

★ IMPORTANT EXAM QUESTION: What is data leakage? Give two examples and prevention methods.

11. Evaluation Strategies

Evaluation measures how well a trained model performs on data that represents the intended deployment setting. The metric should match the task and the cost of different errors.

11.1 Classification Metrics

Metric	Meaning / Formula
Accuracy	(TP + TN) / (TP + TN + FP + FN). Proportion of all predictions that are correct.
Precision	TP / (TP + FP). Of predicted positives, how many are actually positive?
Recall / Sensitivity	TP / (TP + FN). Of actual positives, how many were detected?
Specificity	TN / (TN + FP). Of actual negatives, how many were correctly identified?
F1-score	2 × (Precision × Recall) / (Precision + Recall). Harmonic mean of precision and recall.

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Use-case interpretation: in a disease-screening or safety-related application, missing a real positive (FN) may be costly, so recall can be especially important. In a system where false alarms are costly, precision may receive more emphasis.

11.2 Regression Metrics

Metric	Meaning
MAE	Mean absolute error; easy to interpret in target units.
MSE	Mean squared error; penalizes larger errors more strongly.
RMSE	Square root of MSE; returns to target units.
R^2	Proportion of variance explained relative to a baseline mean model; can be negative on test data.

11.3 Cross-Validation

In **k-fold cross-validation**, the training data is divided into k folds. The model is trained on k–1 folds and validated on the remaining fold. This is repeated k times so every fold serves as validation once. The scores are then summarized, commonly by their mean and variability.

★ IMPORTANT EXAM QUESTION: Explain confusion matrix and derive accuracy, precision, recall and F1-score.

★ IMPORTANT EXAM QUESTION: When is accuracy misleading? Explain with an imbalanced-class example.

★ IMPORTANT EXAM QUESTION: Explain k-fold cross-validation with a neat diagram.

★ IMPORTANT EXAM QUESTION: Compare MAE, MSE, RMSE and R^2.

★ IMPORTANT EXAM QUESTION: Why is cross-validation useful for model evaluation and selection?

Exam Tip: For numerical questions, write TP, TN, FP, FN first. Then substitute values carefully into formulas.

12. Case Study – Customer Retention Prediction Using Behavioral Analytics

Customer retention prediction, often framed as **churn prediction**, is a supervised machine-learning problem. The objective is to predict whether a customer is likely to stop using a service within a defined future period.

Step 1: Define the problem

Target example: **churn = 1** if the customer becomes inactive or cancels within the next 30 days; otherwise **churn = 0**. The prediction horizon and churn definition must be fixed before feature construction.

Step 2: Collect behavioral data

• Login frequency and recency

• Number and value of purchases

• Session duration

• Feature/product usage

• Customer support interactions

• Subscription tenure

• Complaints or service issues

• Payment failures

Step 3: EDA and preprocessing

Check class balance, missing values, duplicate records, unusual values and relationships between behavior and churn. Encode categorical variables, scale numerical features when needed, and handle outliers carefully.

Step 4: Feature engineering

Feature	Interpretation
Recency	Days since last activity/purchase
Frequency	Number of activities or purchases in a defined period
Monetary value	Amount spent in a defined period
Usage trend	Change in usage over time
Support frequency	Number of support interactions

Feature	Interpretation
Payment failure count	Potential friction in subscription/payment

Step 5: Model and evaluation

Possible baseline models include Logistic Regression and Decision Tree; other models can be evaluated depending on the course scope. For an imbalanced churn problem, accuracy alone may be insufficient. Precision, recall, F1-score, ROC-AUC/PR-AUC and the business cost of false positives/negatives can be considered.

Step 6: Business action

Customers predicted as high-risk can be placed into an intervention workflow such as personalized communication, support outreach or a retention offer. The intervention should be evaluated using an appropriate experiment or business KPI rather than assuming the prediction itself causes retention.

★ IMPORTANT EXAM QUESTION: Explain the complete customer retention prediction case study using behavioral analytics.

★ IMPORTANT EXAM QUESTION: What features would you engineer for customer churn prediction and why?

★ IMPORTANT EXAM QUESTION: Which evaluation metrics are useful for churn prediction and why can accuracy be misleading?

13. Practical Focus – What You Should Be Able to Do

A. Data preprocessing practical

• Load a dataset using pandas.

• Inspect shape, columns, data types and missing values.

• Remove or handle duplicates.

• Impute missing numerical/categorical values.

• Encode categorical columns.

• Scale numerical features when the algorithm requires it.

• Separate features X and target y.

• Create train/test or train/validation/test splits.

B. EDA practical

• Generate summary statistics.

• Plot distributions and box plots.

• Inspect target/class balance.

• Calculate correlations where meaningful.

• Identify possible outliers and missingness patterns.

• Form hypotheses from the data before modeling.

C. Model pipeline practical

A clean conceptual pipeline is: **raw data** → **split** → **preprocessing fitted on training data** → **feature transformation** → **model training** → **validation/tuning** → **final test** → **deployment**.

★ IMPORTANT EXAM QUESTION: Explain how you would build an ML workflow from a raw CSV file to final model evaluation.

★ IMPORTANT EXAM QUESTION: Why should preprocessing be included inside a reproducible pipeline?

Exam Tip: If asked for code in the practical exam, explain the purpose of each block rather than only writing syntax.

14. Most Important Exam Questions – Priority List

Prepare these first. The questions are organized by expected conceptual value, not by a prediction of the exact paper.

★★★★★ Very High Priority

★ IMPORTANT EXAM QUESTION: Define Machine Learning and explain its applications and basic terminology.

★ IMPORTANT EXAM QUESTION: Differentiate AI, ML and Deep Learning with a suitable diagram and examples.

★ IMPORTANT EXAM QUESTION: Explain supervised, unsupervised, semi-supervised and reinforcement learning.

★ IMPORTANT EXAM QUESTION: Explain the complete Machine Learning pipeline with all stages.

★ IMPORTANT EXAM QUESTION: Explain data preprocessing: missing values, encoding, scaling, outliers and duplicates.

★ IMPORTANT EXAM QUESTION: What is feature engineering? Explain its major techniques with examples.

★ IMPORTANT EXAM QUESTION: Explain overfitting, underfitting and the bias–variance trade-off.

★ IMPORTANT EXAM QUESTION: Differentiate training, validation and test sets. Explain data leakage.

★ IMPORTANT EXAM QUESTION: Explain confusion matrix and accuracy, precision, recall, specificity and F1-score.

★ IMPORTANT EXAM QUESTION: Explain k-fold cross-validation and its advantages.

★ IMPORTANT EXAM QUESTION: Explain the customer retention prediction case study using behavioral analytics.

★★★★ Important

★ IMPORTANT EXAM QUESTION: What is EDA? Explain its steps and commonly used visualizations.

★ IMPORTANT EXAM QUESTION: Differentiate classification and regression with examples.

★ IMPORTANT EXAM QUESTION: Compare standardization, min-max scaling and robust scaling.

★ IMPORTANT EXAM QUESTION: Explain one-hot encoding and label/ordinal encoding.

★ IMPORTANT EXAM QUESTION: Compare MAE, MSE, RMSE and R^2.

★ IMPORTANT EXAM QUESTION: Why can accuracy be misleading for imbalanced datasets?

★ IMPORTANT EXAM QUESTION: Explain feature selection vs feature extraction.

★ IMPORTANT EXAM QUESTION: Explain model parameters and hyperparameters with examples.

★ IMPORTANT EXAM QUESTION: Explain data leakage with two examples and prevention methods.

15. Short Answer / 2–3 Mark Question Bank

1. What is a feature?

2. What is a target variable?

3. Define supervised learning.

4. Define unsupervised learning.

5. What is classification?

6. What is regression?

7. What is clustering?

8. What is overfitting?

9. What is underfitting?

10. Define bias and variance.

11. What is feature scaling?

12. What is one-hot encoding?

13. What is an outlier?

14. What is EDA?

15. What is cross-validation?

16. What is data leakage?

17. What is a hyperparameter?

18. What is a confusion matrix?

19. Define precision and recall.

20. What is the purpose of a validation set?

21. Why is the test set kept untouched?

22. What is feature engineering?

23. What is model generalization?

24. Why is preprocessing important?

16. Formula & Definition Sheet for Last-Minute Revision

Concept	Remember
Standardization	$z = (x - \mu) / \sigma$
Min-Max	$x' = (x - \min) / (\max - \min)$
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
F1	$2PR / (P + R)$
MAE	Mean(|y – ■|)
MSE	Mean((y – ■)²)
RMSE	√MSE
R^2	$1 - SS_{res} / SS_{tot}$
IQR	Q3 – Q1
AI–ML–DL	AI ⊃ ML ⊃ DL
Pipeline	Problem → Data → EDA → Preprocess → Features → Split → Train → Validate → Test → Deploy → Monitor

17. How to Write High-Scoring Theory Answers

• **Start with a precise definition.** Do not begin with a vague statement.

• **Use headings.** For example: definition, working, advantages, limitations, example.

• **Use comparison tables** for AI vs ML vs DL, supervised vs unsupervised, train vs validation vs test, etc.

• **Draw diagrams** for ML pipeline, bias–variance, confusion matrix and k-fold cross-validation when appropriate.

• **Give a practical example.** Examples make abstract concepts easier to evaluate.

• **Write formulas accurately** and define symbols such as TP, FP, FN and TN.

• **For case studies, follow a workflow:** problem → data → features → preprocessing → model → evaluation → business action.

• **Do not confuse correlation with causation.** A predictive feature does not automatically cause the target.

Exam Tip: For a 10-mark answer: Definition (1) + core explanation (3) + diagram/table (2) + example (2) + advantages/limitations or conclusion (2). Adapt to your university's marking scheme.

18. One-Day / One-Hour Revision Strategy

Time	Revise
10 min	AI vs ML vs DL + types of learning
10 min	ML pipeline + preprocessing
10 min	Feature engineering + EDA
10 min	Overfitting + underfitting + bias–variance
10 min	Train/validation/test + leakage + cross-validation
10 min	Confusion matrix + formulas + customer retention case study

Final checklist: If you can explain every ★★★★★ question without looking at the notes, draw the ML pipeline, derive the four confusion-matrix metrics, explain bias–variance, and describe the customer-retention case study end-to-end, you have covered the core exam material of this unit.

End of Unit 1 – Machine Learning Foundations and Statistical Learning